

A PERTURBATIVE SOLUTION OF THE TRANSFER EQUATIONS FOR THE STOKES PARAMETERS IN A MAGNETIC FIELD

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Abstract. The transfer equations for the Stokes parameters, as derived under the assumption of LTE and in the general case of anomalous Zeeman multiplet, are solved by a perturbative method that reduces the system of four coupled linear differential equations to an iterative series of well known linear differential equations. The perturbation parameter turns out to be of the order of magnitude of the ratio of the Larmor frequency to the line Doppler half-width, so that the method appears to be particularly suitable for small magnetic fields.

Some qualitative results on the importance of Faraday rotation and the orders of magnitude of the line profiles of the Stokes parameters are obtained.

1. Introduction

The radiative transfer equations for polarized light, under the hypothesis of LTE, firstly obtained by Unno (1956), have been subsequently generalized by various authors to account for anomalous dispersion and for the depth dependence of the azimuth of the magnetic field vector (Rachkovsky, 1962, 1967; Beckers, 1969a, b; Stenflo, 1971; Landi Degl'Innocenti and Landi Degl'Innocenti, 1972). The calculation of line profiles for the Stokes parameters requires then the knowledge of the solution of these equations. This has been the task of various authors (Hubenet, 1955; Unno 1956; Kjeldseth Moe, 1968; Michard, 1961; Mattig, 1966). In fact they gave a solution of the equations under some restrictive assumptions about: (a) the depth dependence of the magnetic field; (b) the atmosphere model; (c) the depth dependence of the Voigt-function profile of the line. We refer the reader to the paper by Göhring (1971) for a detailed review on the various assumptions introduced by these authors.

If we want to account for a realistic model of the solar atmosphere and the depth dependence of the magnetic field (both in magnitude and direction), the formal solutions we mentioned above, are not directly applicable and we must turn our attention to more sophisticated methods such as those of Staude (1969), Beckers (1969a), Göhring (1971). The method of Staude solves the transfer equations for arbitrary depth dependence of all physical parameters; nevertheless, as the same author says (Staude, 1970), it has first of all a theoretical significance as the solution has a too complicated formal structure. On the other hand, the numerical methods of Beckers and Göhring, though fully satisfactory from a mathematical viewpoint, have the disadvantage of being not too easy to handle especially if one is interested in an estimate of order of magnitude rather than in an accurate calculation.

The method we propose, though quickly converging only for small magnetic fields, is very simple and has the advantage of disentangling the equations in such a way that linear differential equations of the same type as those of the standard theory of radiative transfer, for which a formal solution is already known, are obtained. Moreover, from this solution, we can derive some very important qualitative consequences for the theory of line formation in a magnetic field and magnetographs calibration.

2. Theory

The equations of radiative transfer we refer to, are those derived in our preceding paper (1972) (hereafter referred as Paper I)

$$\begin{aligned}\cos \alpha \frac{dI}{d\tau} &= (1 + \eta_I) (I - B) + \eta_Q Q + \eta_U U + \eta_V V, \\ \cos \alpha \frac{dQ}{d\tau} &= \eta_Q (I - B) + (1 + \eta_I) Q + \varrho_V U - \varrho_U V, \\ \cos \alpha \frac{dU}{d\tau} &= \eta_U (I - B) + (1 + \eta_I) U - \varrho_V Q + \varrho_Q V, \\ \cos \alpha \frac{dV}{d\tau} &= \eta_V (I - B) + (1 + \eta_I) V + \varrho_U Q - \varrho_Q U,\end{aligned}\tag{1}$$

where we have changed by a factor $\sqrt{2}$ the definitions of U and V , in order to adapt our definition to the operational one given by Shurcliff (1962), and where all the other notations have the same meaning.

Let us now introduce a real parameter λ , which at the end of the calculations will be set equal to 1, and let us think of the magnetic field at the various depths τ , as if it were λ times the real magnetic field just present in the τ -depth lineforming layers. That is we make the transformation

$$\mathbf{B}(\tau) \rightarrow \lambda \mathbf{B}(\tau).$$

By so doing all the quantities in the Equations (1) become functions of λ and they can be expanded in power series; we get so for the Stokes parameters

$$I(\lambda) = I_0 + I_1 \lambda + I_2 \lambda^2 + I_3 \lambda^3 + \dots$$

and analogous expressions for Q , U , V .

For η_p , η_b , η_r , in the general case of anomalous Zeeman splitting, we have the following expansions (up to the third order)

$$\begin{aligned}\eta_p &= \eta + \frac{1}{2} \eta'' v_L^2 G_0 + O(v_L^4), \\ \eta_{b,r} &= \eta \mp \eta' v_L G_1 + \frac{1}{2} \eta'' v_L^2 G_2 \mp \frac{1}{6} \eta''' v_L^3 G_3 + O(v_L^4),\end{aligned}\tag{2}$$

where η is simply given by

$$\eta = \frac{\chi}{k_v^{(c)}} \mathcal{H}(v; v_0, \gamma, T),$$

i.e. the Voigt function centered at the frequency ν_0 , of the unperturbed line; η' , η'' , η''' , are the derivatives of η with respect to the frequency; $\nu_L = e_0 B(\tau)/4\pi mc$ is the Larmor frequency corresponding to the magnetic field B , and contains the parameter λ linearly; finally G_0 , G_1 , G_2 , G_3 , are given by the expressions

$$\begin{aligned} G_0 &= (g' - g)^2 \Sigma_3^2, \\ G_1 &= g' + (g' - g) \Sigma_1^1, \\ G_2 &= g'^2 + 2g'(g' - g) \Sigma_1^1 + (g' - g)^2 \Sigma_1^2, \\ G_3 &= g'^3 + 3g'^2(g' - g) \Sigma_1^1 + 3g'(g' - g)^2 \Sigma_1^2 + (g' - g)^3 \Sigma_1^3, \end{aligned}$$

where g and g' are the Landé factors of the lower and upper levels respectively and where the various quantities Σ_i^k are defined in the following manner:

$$\Sigma_i^k = \sum_M S_i^{(M)} M^k \quad (i = 1, 2, 3),$$

while the quantities $S_i^{(M)}$ are defined as in Paper I. The values of the quantities $S_i^{(M)}$ depend on the quantum numbers J and J' of the angular momentum of the lower and the upper level and are given by Condon and Shortley (1967); we obtain for the first non-zero Σ_i^k the values reported in Table I.

TABLE I
Values of Σ_i^k for the transition $J \rightarrow J'$

	$J' = J + 1$	$J' = J$	$J' = J - 1$
$\Sigma_1^1 = -\Sigma_3^1$	$J/2$	$-1/2$	$-(J+1)/2$
Σ_3^2	$J(J+2)/5$	$(3J^2+3J-1)/5$	$(J+1)(J-1)/5$
$\Sigma_1^2 = \Sigma_2^2$	$J(4J+3)/10$	$(2J^2+2J+1)/10$	$(J+1)(4J+1)/10$
$\Sigma_1^3 = -\Sigma_2^3$	$J(3J^2+3J-1)/10$	$-(3J^2+3J-1)/10$	$-(J+1)(3J^2+3J-1)/10$

The quantities G_0 , G_1 , G_2 , G_3 have very simple expressions in some particular cases; in the case of normal Zeeman triplet they are

$$G_0 = 0 \quad G_1 = g \quad G_2 = g^2 \quad G_3 = g^3, *$$

and in the case of a transition $J = \frac{1}{2} \rightarrow J' = \frac{1}{2}$ they are

$$G_1 = \bar{g} \quad G_2 = \bar{g}^2 \quad G_3 = \bar{g}^3 \quad \text{where} \quad \bar{g} = \frac{g + g'}{2}.$$

For q_p , q_b , q_r , we obtain expressions similar to (2).

Substituting Equations (2) in the expressions for η_I etc., we obtain

$$\begin{aligned} \eta_I &= \eta + \frac{1}{4}\eta''v_L^2[G_0 \sin^2 \theta + G_2(1 + \cos^2 \theta)] + O(v_L^4), \\ \eta_Q &= \frac{1}{4}\eta''v_L^2(G_0 - G_2) \sin^2 \theta \cos 2\varphi + O(v_L^4), \\ \eta_U &= \frac{1}{4}\eta''v_L^2(G_0 - G_2) \sin^2 \theta \sin 2\varphi + O(v_L^4), \\ \eta_V &= \eta'v_L G_1 \cos \theta + \frac{1}{8}\eta'''v_L^3 G_3 \cos \theta + O(v_L^5). \end{aligned} \tag{3}$$

* In the transition $J = 0 \rightarrow J' = 1$, g must be substituted by g' .

$\varrho_Q, \varrho_U, \varrho_V$, are the same as η_Q, η_U, η_V , with the substitutions

$$\eta', \eta'', \eta''' \rightarrow \varrho', \varrho'', \varrho'''.$$

Let us now turn our attention to the boundary conditions of Equations (1). Since the boundary values for I, Q, U, V are the following

$$I(\tau) \xrightarrow{\tau \rightarrow \infty} B(\tau),$$

$$Q(\tau) \xrightarrow{\tau \rightarrow \infty} 0,$$

$$U(\tau) \xrightarrow{\tau \rightarrow \infty} 0,$$

$$V(\tau) \xrightarrow{\tau \rightarrow \infty} 0,$$

and since they must hold for every value of the λ parameter, we have

$$I_0(\tau) \xrightarrow{\tau \rightarrow \infty} B(\tau),$$

and for all the other quantities $I_1, I_2, I_3, \dots; Q_0, Q_1, \dots; U_0, U_1, \dots; V_0, V_1, \dots$ we have 0 as boundary value.

Substituting expressions (3) in Equations (1) and equating the various powers of λ we obtain at various orders: 'zero order':

$$\begin{aligned} \cos \alpha \frac{dI_0}{d\tau} &= (1 + \eta) (I_0 - B); & \cos \alpha \frac{dQ_0}{d\tau} &= (1 + \eta) Q_0; \\ \cos \alpha \frac{dU_0}{d\tau} &= (1 + \eta) U_0; & \cos \alpha \frac{dV_0}{d\tau} &= (1 + \eta) V_0; \end{aligned} \quad (4a)$$

taking into account the boundary conditions, we see that the last three equations are satisfied by the solutions: $Q_0 = U_0 = V_0 = 0$.

In the same way we obtain for the higher orders: 'first order':

$$\begin{aligned} I_1 &= Q_1 = U_1 = 0, \\ \cos \alpha \frac{dV_1}{d\tau} &= (1 + \eta) V_1 + \eta' v_L G_1 \cos \theta (I_0 - B), \end{aligned} \quad (4b)$$

'second order':

$$\begin{aligned} \cos \alpha \frac{dI_2}{d\tau} &= (1 + \eta) I_2 + \frac{1}{4} \eta'' v_L^2 [G_0 \sin^2 \theta + G_2 (1 + \cos^2 \theta)] \times \\ &\quad \times (I_0 - B) + \eta' v_L G_1 \cos \theta V_1, \\ \cos \alpha \frac{dQ_2}{d\tau} &= (1 + \eta) Q_2 + \frac{1}{4} \eta'' v_L^2 (G_0 - G_2) \sin^2 \theta \cos 2\varphi (I_0 - B), \\ \cos \alpha \frac{dU_2}{d\tau} &= (1 + \eta) U_2 + \frac{1}{4} \eta'' v_L^2 (G_0 - G_2) \sin^2 \theta \sin 2\varphi (I_0 - B), \\ V_2 &= 0, \end{aligned} \quad (4c)$$

'third order':

$$\begin{aligned}
 I_3 &= 0 \quad , \\
 \cos \alpha \frac{dQ_3}{d\tau} &= (1 + \eta) Q_3 + \varrho' v_L G_1 \cos \theta U_2 - \frac{1}{4} \varrho'' v_L^2 (G_0 - G_2) \sin^2 \theta \sin 2\varphi V_1, \\
 \cos \alpha \frac{dU_3}{d\tau} &= (1 + \eta) U_3 - \varrho' v_L G_1 \cos \theta Q_2 + \frac{1}{4} \varrho'' v_L^2 (G_0 - G_2) \sin^2 \theta \cos 2\varphi V_1, \\
 \cos \alpha \frac{dV_3}{d\tau} &= (1 + \eta) V_3 + \frac{1}{8} \eta''' v_L^3 G_3 \cos \theta (I_0 - B) + \\
 &\quad + \eta' v_L G_1 \cos \theta I_2 + \frac{1}{4} \eta'' v_L^2 [G_0 \sin^2 \theta + G_2 (1 + \cos^2 \theta)] V_1, \\
 &\text{etc.}
 \end{aligned}$$

3. Discussion and Conclusions

As can be easily seen from the equations and the boundary conditions, I_0 is just the intensity of the radiation field that one would have if the magnetic field were not present. The first non-zero correction to the value of the Stokes parameters is V_1 and it is of the order of magnitude of $x = v_L / \Delta v_D$; the corrections which arise from the n th order are, always in order of magnitude, of order x^n .

From the equations we have derived, we are driven to the following considerations:

(i) The Faraday rotation, represented in Equations (1) by the terms ϱ_Q , ϱ_U , ϱ_V , has effect only of order x^3 on Q and U , but has no influence until the fourth order on I and V . In other words, if we don't take into account the Faraday rotation, we make an error of order x on U and Q , of order x^3 on V and of order x^4 on I^* ; so, if the magnetic field is sufficiently small so that $x \approx 0.1$, the Faraday rotation may be neglected in Equations (1) if we are not interested in a precision greater than 10%. On the other hand we can say that if we measure the quantity V with a longitudinal magnetograph, the effect of Faraday rotation may be neglected if we do not consider too large magnetic fields.

(ii) The Stokes parameter, which differs at a lower order from its value without magnetic field, is V ; it is particularly suitable to measure the longitudinal component of the magnetic field, as in the differential equation for V_1 the non homogeneous term contains the factor $v_L \cos \theta$ which is proportional to the quantity just mentioned.

(iii) For a similar reason Q and U , as is already known, are the most suitable Stokes parameters for the study of the transversal component of the magnetic field.

(iv) Considering the v -derivative of the Equation (4a) for I_0 , we obtain**

$$\cos \alpha \frac{d}{d\tau} \frac{dI_0}{dv} = (1 + \eta) \frac{dI_0}{dv} + \eta' (I_0 - B),$$

and if we suppose that the quantity $v_L \cos \theta$ be constant along the line of sight, i.e.

* In some particular cases, e.g. for $\theta = \text{const.} = 0$, some quantities as U_2 , or Q_2 , may be zero so that this conclusion may not hold.

** The v -derivative of B_v may be neglected in the line.

that the magnetic field has always a constant component along this line, we can write

$$\cos \alpha \frac{d}{d\tau} \left(G_1 v_L \cos \theta \frac{dI_0}{dv} \right) = (1 + \eta) G_1 v_L \cos \theta \frac{dI_0}{dv} + \eta' G_1 v_L \cos \theta (I_0 - B).$$

Comparing this equation with the Equation (4b) for V_1 , we see that the two quantities V_1 and $G_1 v_L \cos \theta (dI_0/dv)$ obey the same differential equation; moreover they have the same boundary condition, so that we can state that at every τ and for every α

$$V_1 = G_1 v_L \cos \theta \frac{dI_0}{dv},$$

and in particular at $\tau = 0$

$$V_1(0) = G_1 v_L \cos \theta \frac{dI_0(0)}{dv}.$$

This means that, *to the first order in x* , V is proportional to the derivative of the intensity I_0 .

(v) Under the same assumption of the constance of the component of the magnetic field along the line of sight, if the following condition holds:

$$\tan^2 \theta = \frac{2(G_1^2 - G_2)}{G_0 + G_2}, \quad (5)$$

we obtain in a similar manner

$$I_2 = \frac{1}{2} v_L^2 G_1^2 \cos^2 \theta \frac{d^2 I_0}{dv^2}.$$

A careful analysis of the condition (5) shows that in cases of normal Zeeman triplet it can be satisfied only for $\theta = 0$, and that in the cases of anomalous Zeeman splitting it is never satisfied except for the transition $J = \frac{1}{2} \rightarrow J' = \frac{1}{2}$ with $\theta = 0$.

We have, when the condition is satisfied,

$$I_2 = \frac{1}{2} v_L^2 G_1^2 \frac{d^2 I_0}{dv^2}.$$

Analogously we can show that, if the condition (5) holds, the quantity V_3 is given by

$$V_3 = \frac{1}{6} v_L^3 G_1^3 \frac{d^3 I_0}{dv^3}.$$

(vi) If the azimuth φ of the magnetic field is constant, if we multiply the Equation (4c) for Q_2 by $\sin 2\varphi$ and the Equation (4c) for U_2 by $\cos 2\varphi$, we obtain by subtraction

$$\cos \alpha \frac{d}{d\tau} (\sin 2\varphi Q_2 - \cos 2\varphi U_2) = (1 + \eta) (\sin 2\varphi Q_2 - \cos 2\varphi U_2),$$

which gives after integration

$$\sin 2\varphi Q_2 - \cos 2\varphi U_2 = 0,$$

that is

$$\tan 2\varphi = \frac{U_2}{Q_2}.$$

Under the same assumption we can also show that

$$\tan 2\varphi = -\frac{Q_3}{U_3}.$$

(vii) Another interesting relation concerning Q_2 and U_2 can be obtained if the magnetic field is constant:

$$Q_2(v_0) = \frac{1}{4}v_L^2(G_0 - G_2) \sin^2 \theta \cos 2\varphi \left(\frac{d^2 I_0}{dv^2} \right)_{v=v_0},$$

$$U_2(v_0) = \frac{1}{4}v_L^2(G_0 - G_2) \sin^2 \theta \sin 2\varphi \left(\frac{d^2 I_0}{dv^2} \right)_{v=v_0}.$$

All these conclusions are direct consequences of our method of solution. We must remark that some of them have already been obtained by other authors using different methods; e.g.: (i) was obtained by Göhring (1970), (ii)–(iv) by Unno (1956), Nikulin *et al.* (1958), and Severny and Stepanov (1962).

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